

High Speed 5-Bit Carry-Lookahead Adder

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Abstract—This project presents the design and implementation of a high-speed, low-power 4-bit Manchester Carry Chain (MCC) adder. The MCC architecture leverages the inherent propagation efficiency of carry-chain logic, enabling significantly faster carry computation compared to conventional ripple or lookahead schemes. By integrating optimized static CMOS gates and carefully tuned transistor sizing, the proposed MCC achieves substantial improvement in critical-path delay, resulting in a more efficient arithmetic unit. Post-layout simulation and FPGA testing validate the speed and energy benefits of the design, demonstrating a reduced power-delay product and robust performance across operating conditions.

THE AIM OF THIS PROJECT

The aim of this project is to propose a circuit design, which takes two 5-bit numbers as inputs, along with an input carry and computes the sum and output carry. A clock is also given as input to the circuit. The input is taken into the circuit at the rising edge of the clock and the output is computed and reported at the next clock edge.

Each output sum bit needs to drive an inverter of size $W_p/W_n = 20\lambda/10\lambda$, where $\lambda = 0.09 \mu m$.

I. INTRODUCTION

Adders form the foundation of most arithmetic operations in a processor, so their speed has a major impact on overall system performance. Even small delays become significant when an adder is used repeatedly in complex logic.

Basic structures like ripple-carry adders are compact but slow because the carry must propagate through each stage. Faster architectures improve this by reducing carry-propagation time.

This project focuses on using a more efficient carry-generation method to achieve lower delay and power-delay product (PDP), while keeping the area and power overhead minimal.

II. PROPOSED CLA CIRCUIT

The circuit given below shows the classic Manchester Carry Lookahead adder

The principle equations of a CLA are:

$$P_i = A_i \oplus B_i \quad (1)$$

$$G_i = A_i \cdot B_i \quad (2)$$

$$C_i = G_{i-1} + P_{i-1}C_{i-1} \quad (3)$$

$$S_i = P_i \oplus C_i \quad i = 1, 2, 3, 4 \quad (4)$$

Using the above equations, we precompute the carry expressions:

$$C_1 = G_0 + P_0C_0 \quad (5)$$

$$C_2 = G_1 + P_1G_0 + P_1P_0C_0 \quad (6)$$

$$C_3 = G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0C_0 \quad (7)$$

$$C_4 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0C_0 \quad (8)$$

$$C_5 = G_4 + P_4G_3 + P_4P_3G_2 + P_4P_3P_2G_1 + P_4P_3P_2P_1G_0 + P_4P_3P_2P_1P_0C_0 \quad (9)$$

The total critical path delay of the MCLA is:

$$\text{Delay} = D_{\text{PTL-p0}} + D_{\text{PTL-p1}} + D_{\text{PTL-p2}} + D_{\text{PTL-p3}} + D_{\text{PTL-p4}}$$

This represents the worst-case delay, which occurs when all $P_i = 1$ and all $G_i = 0$. In this condition, the discharge path for C_{out} extends all the way back to C_{in} .

III. SUBCIRCUIT BLOCK SPECIFICATIONS

The modified clocked CLA requires many building blocks. Minimum delay occurred when sizing an inverter as $W_p/W_n = 20\lambda/10\lambda$, which in turn occupies lesser area.

Preferring Manchester Carry over static CMOS logic is mainly due the reduced area and more controlled power consumption due to dynamic nature, however, it is not much better than the static CMOS implementation.

A. Flip-Flops

True Single Phase Clock(TSPC) was used. TSPC was chosen due to lower area and power.

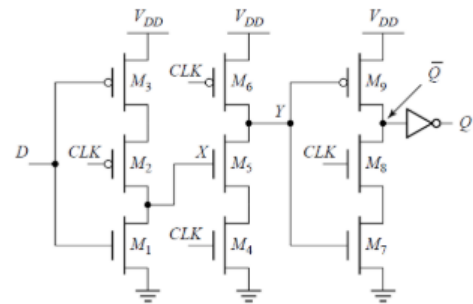


Fig. 1: Conventional TSPC flip-flop

To store the input and output of the adder, TSPC flipflop is used. TSPC is high speed single clock requiring flipflop, which requires lesser area while layouting and also have less complex logic.. The size of PMOS transistors used in this implementation is double the size of NMOS transistors

to equalize the current driving capabilities. Various sizes of transistors were tested to find the configuration with the minimum delay. The minimum delay was found to occur when, the NMOS width is set to 10λ , while the PMOS width is set to 20λ .

B. Adder Block

A Manchester Carry adder block is fairly simple dynamic logic adder.

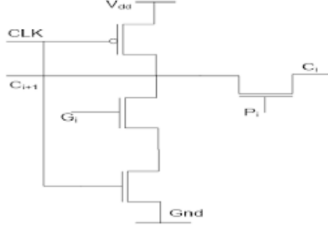


Fig. 2: 1bit Manchester Carry Adder

Since $C_{out} = G_i + P_i C_{i-1}$, we can implement this stage using dynamic CMOS logic as shown in the figure.

When $clk = 0$, C_{out} precharges to 1.

When $clk = 1$:

- If $P = 1$ and $G = 0$, then C_{out} will be evaluated according to C_{in} .
- If $P = 0$ and $G = 1$, then C_{out} will discharge to 0, indicating that the carry-out for this bit will be 1 regardless of C_{in} .

This is how carry can be predicted for some values of the input. Also since the C_{out} bar is precharged for $clk = 0$, C_{out} need not change for values of input for which the output carry is 0, hence faster response is achieved.

To create this adder block, the following gates are used:

- 1) Inverter: NMOS width = 20λ and PMOS width = 10λ .
- 2) 2-input XOR gate: The XOR is implemented using pass-transistor logic (PTL). Since PTL uses only NMOS transistors, an XNOR function is generated first, which is then passed through an inverter to obtain the correct XOR output with full voltage swing.

XOR gate is required for logic of P, However logic of G can be further simplified. As $G = AB$, we can replace the nmos whose input is G, by two nmos transistors having input A and B in series. The function of the circuit is same, and requiring lesser area as NAND gate can be fully bypassed.

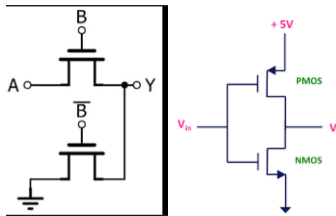


Fig. 3: basic XOR NMOS PTL and an inverter

IV. NGSPICE SIMULATIONS

A. Flip Flop

A positive edge D flipflop is made using TSPC logic

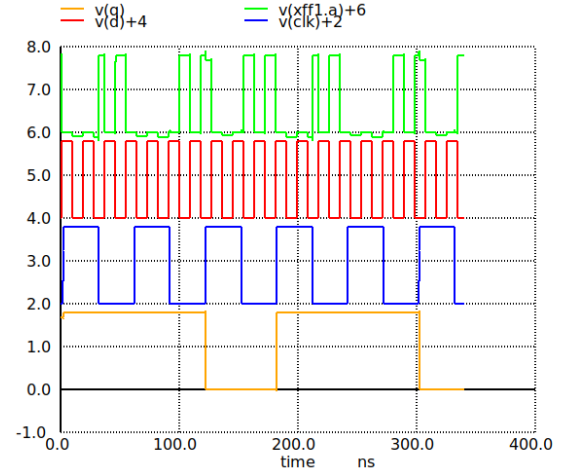


Fig. 4: Waveform for D TSPC Flipflop
($xff1.a$ refers to the pt X in the circuit diagram)

- 1) Setup Time: It is the minimum time before the rising edge of clock for which the input must be stabilised. If the input is changed in this time, the value of the intermediate state is changed, hence the output cannot be properly latched, leading to errors. The setup time can be calculate as the time delay between D and X(as can be seen in the circuit diagram of TSPC). The value of setup time is - 56.6437ps
- 2) Hold Time: It is the minimum time the input must be stabilized before it can be changed after the rising edge of the clock. Generally, it is the delay required between the input to output propgation, but since in a flipflop, half of the propgation(D to X) happens when $clk = 0$, and the later half of the propgation(X to Q) happens when $clk = 1$, enables this logic to have a hold time of zero.
- 3) Clk to Q delay: Since Q changes only on the poisitve edge of the clk , it doesnt make sense to calculate the delay between the input(D) change and the output(Q) change. Rather the delay between the rising edge of the clock and the change of Q is important as it gives us the essential delay of the flipflop. This delay comes out to be about 140.84ps.

q_delay_fall	=	1.702383e-10	targ=	1.222202e-07	trig=	1.220500e-07
q_delay_rise	=	1.114561e-10	targ=	1.821615e-07	trig=	1.820500e-07
tp2q	=	1.408472e-10				
tsu_r	=	4.150199e-11	targ=	5.509150e-08	trig=	5.505000e-08
tsu_f	=	7.178560e-11	targ=	4.622179e-08	trig=	4.615000e-08
tsu	=	5.664379e-11				

Fig. 5: FlipFlop delay and setup time

B. Manchester Carry Chain

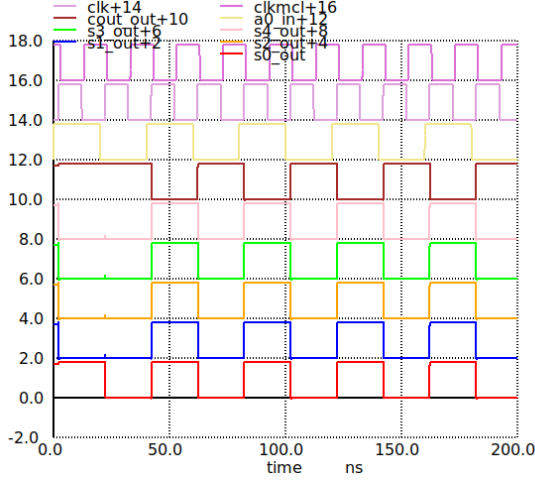


Fig. 6: 5bit Manchester Carry Adder with flipflops at input and output

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VA0 A0_in gnd PULSE(0 {SUPPLY} 0 100p 100p 20n 40n)
VB0 B0_in gnd 0
VA1 A1_in gnd 1.8
VB1 B1_in gnd 0
VA2 A2_in gnd 0
VB2 B2_in gnd 1.8
VA3 A3_in gnd 1.8
VB3 B3_in gnd 0
VA4 A4_in gnd 1.8
VB4 B4_in gnd 0
VCin_raw Cin_in gnd 1.8

```

Fig. 7: The above waveforms are for this particular input

This is the Manchester carry chain, represents the carry propagation of 5-bit adder. As mentioned earlier, the maximum delay occurs when Coutbar changes from 1 to 0, when all the propagates are 1, as this time Coutbar will have to discharge at Cin.(longest path has the greatest delay). This delay is calculated and comes out to be - It is to be noted that this is the worst case delay and may not occur everytime. Therefore the delay of the manchester carry chain is variable depending on the inputs, as opposed to normal static cmos implementation.

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dcarrymax = 7.691196e-10 targ= 3.091912e-08 trig= 3.015000e-08

```

Fig. 8: Maximum delay of the manchester carry chain will be

In the adder circuit, the Manchester carry chain is used to precompute the carry whenever and wherever possible. In addition, the sum is computed using

$$S_i = P_i \oplus C_{i-1}.$$

So another XOR gate is required to compute this, however the delay of XOR is minimum and comes out to be -

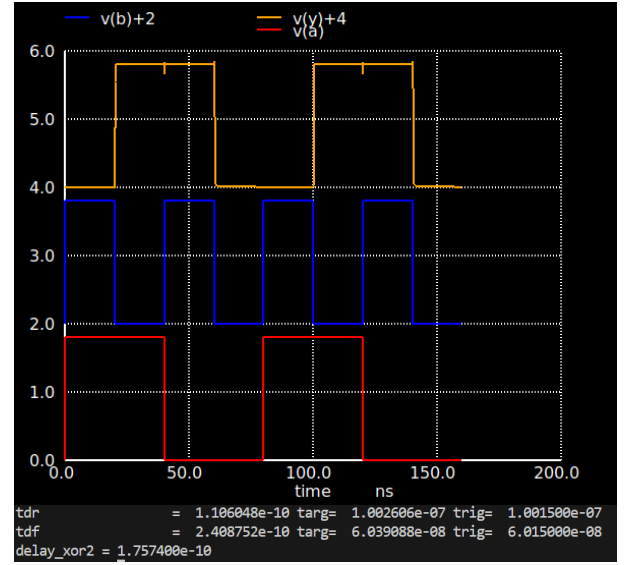


Fig. 9: XOR PTL logic waveform and delay

C. Inverter:

Last but not least, static CMOS inverter is always required in CMOS implementation due to the inverting logic concept. Hence the most basic and vastly used piece of circuit. Width of NMOS is $10 \times \lambda$ and width of PMOS is $20 \times \lambda$. Delay of the inverter is 29.175ps

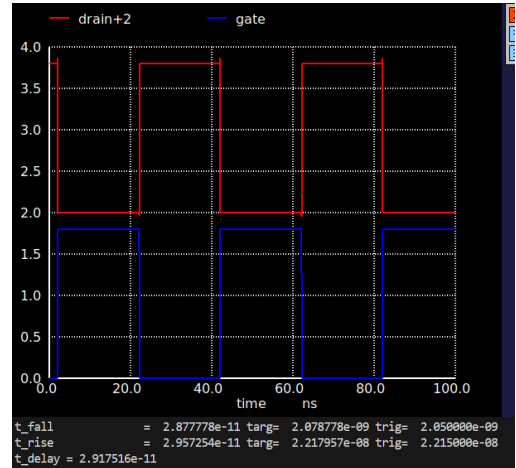


Fig. 10: Waveform and delay of the inverter

D. Timing analysis of the CLK

Since only a single clock is available initially to drive the flip-flops, an inverter is used to generate a complementary and slightly skewed version of this clock. Let these two clocks be denoted as clk_{ff} and clk_{mcl} , respectively. The skew between them is defined as θ , which represents the time difference between the rising edge of clk_{mcl} and the rising edge of clk_{ff} . Let T be the clock period.

To ensure correct operation, two conditions must be satisfied:

1. On every rising edge of the main clock, the new input must be captured by the input flip-flops, and the output corresponding to the previous input must be transferred to the output flip-flops. This requires

$$\theta > D_{mcl} + t_{su,ff},$$

where D_{mcl} is the maximum delay of the Manchester carry chain and $t_{su,ff}$ is the setup time of the flip-flops.

2. For the output flip-flops to correctly store the evaluated result, the following condition must hold:

$$\theta + D_{ff} < \frac{T}{2}.$$

Here D_{ff} is the time required for the output Q to be evaluated after the rising edge of the clock, the condition becomes

This relationship determines the minimum allowable clock period for reliable operation of the circuit, which is to have the sum of A and B by the next positive edge of the clock.

Using the two timing constraints and the measured values $D_{ff} = 140.8$ ps, $t_{su,ff} = 56.64$ ps, and $D_{mca} = 769$ ps, the total required time is

$$D_{ff} + t_{su,ff} + D_{mca} = 140.8 + 56.64 + 769 = 966.44 \text{ ps}.$$

Therefore, the clock period must satisfy

$$T > 2 \times 966.44 \text{ ps} = 1.932 \text{ ns}.$$

Hence, the maximum allowable clock frequency is

$$f_{\max} = \frac{1}{T} = \frac{1}{1.932 \text{ ns}} \approx 0.834 \text{ GHz}.$$

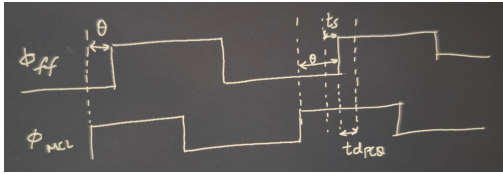


Fig. 11: Tclk Analysis

V. VERILOG SIMULATIONS

Verilog has been simulated for certain fixed inputs only, can be easily varied as required.

A. D Flip Flop

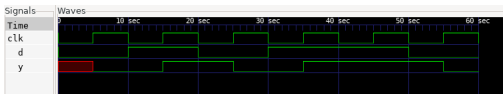


Fig. 12: D flipflop simulation on Verilog

B. 1 Bit Manchester Carry

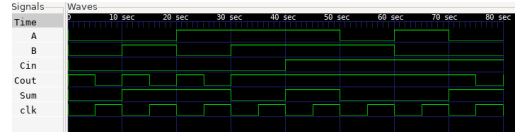


Fig. 13: Verilog Simulation of 1Bit MCA, without flipflops

C. 5 Bit Manchester Carry

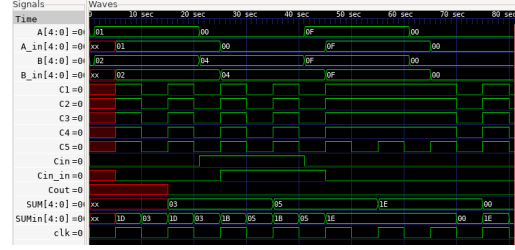


Fig. 14: 5bit MCA with input and output Flipflops

VI. FPGA SIMULATIONS

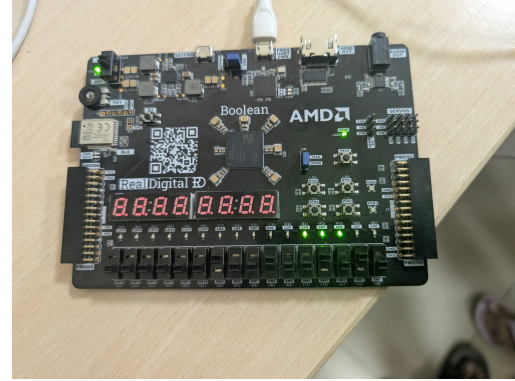


Fig. 15: Image1 FPGA

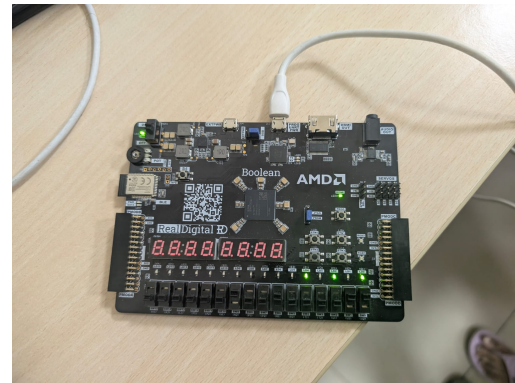


Fig. 16: Image2 FPGA

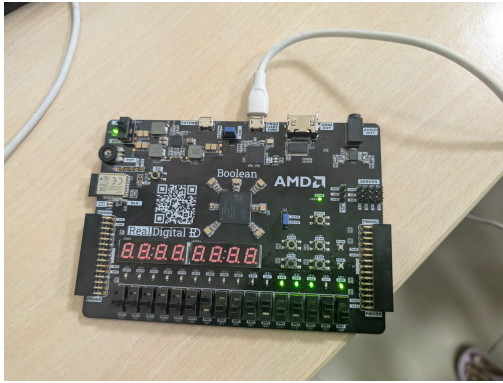


Fig. 17: Image3 FPGA

The inputs are the switches below, right to left they are from A0...A4, B0...B4, Cin. the outputs are the LEDs from the right to left S0...S5, Cout.

When the outputs are plotted on oscilloscope for a 1 bit adder Sum is the green wave and Cout is the yellow wave -

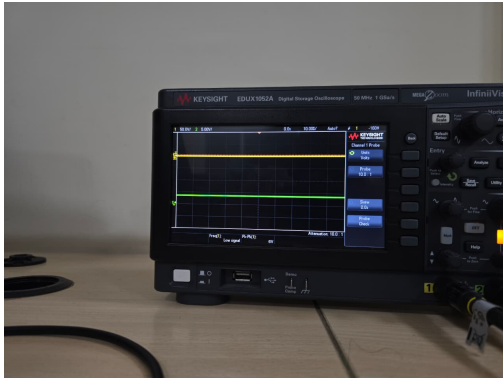


Fig. 18: Sum = 1 , Carry = 0

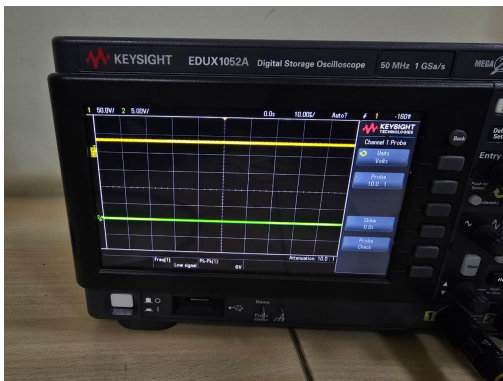


Fig. 19: Sum = 0 , Carry 1



Fig. 20: Sum = 1 , Carry = 1

VII. MAGIC SIMULATIONS

A. Flipflop Layout

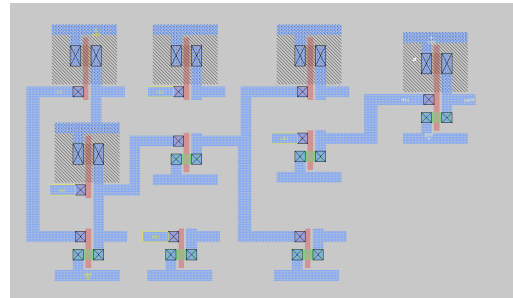


Fig. 21: FlipFlop Layout

B. XOr Gate

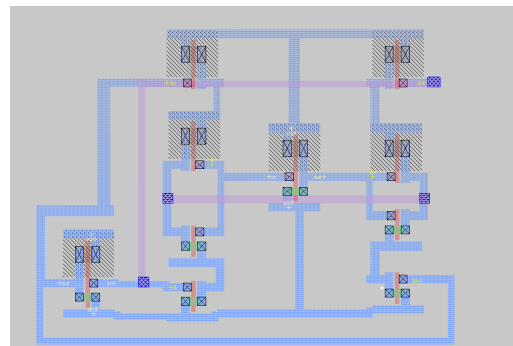


Fig. 22: Xor gate Layout

C. Mca-1bit

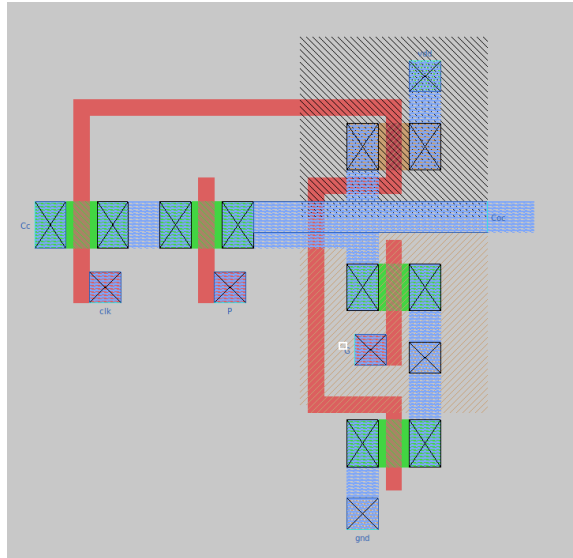


Fig. 23: MCA- 1bit layout

D. Mca-5bit

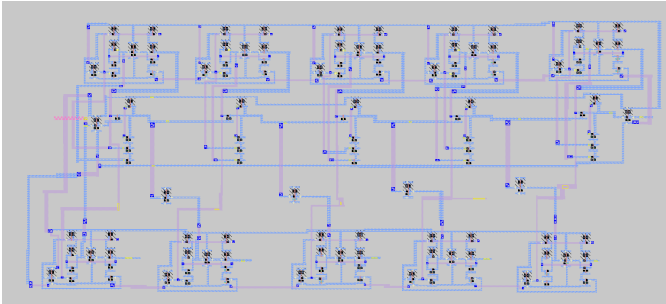


Fig. 24: 5bit Manchester carry lookahead adder without input and output flipflops

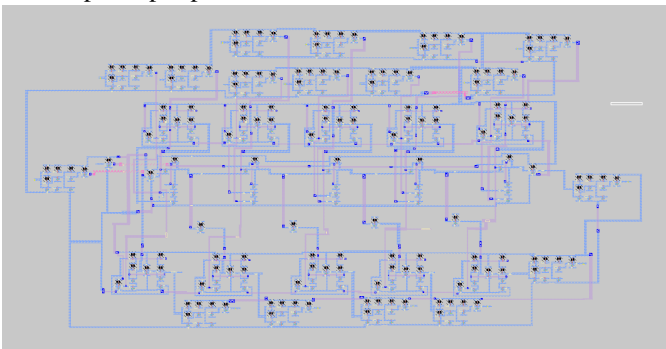


Fig. 25: 5bit Manchester carry lookahead adder with flipflops for input and output

Total Length of the layout is around 199.8um Total width of the layout is around 97.2um Therefore total area required = 19420 μm^2